

[3]

Variability of unsaturated zone water transport parameters: implications for hydrological modelling.

1. In situ measurements

R. Ragab*, J.D. Cooper

Institute of Hydrology, Wallingford, OX10 8BB, UK

(Received 27 September 1992; revision accepted 19 February 1993)

Abstract

Physically based hydrological models require values of saturated and unsaturated hydraulic conductivity to solve the equations used to describe subsurface flow. Variability of these parameters within and between various land uses and soil types is an important consideration if hydrological behaviour is to be described realistically.

A catchment in south-west England has been chosen to validate the European Hydrological System (Système Hydrologique Européen — SHE). The catchment includes three land use domains — arable (including temporary grass), permanent grassland and woodland. For the unsaturated zone component of the SHE both saturated hydraulic conductivity, K_s , and unsaturated hydraulic conductivity as a function of matric suction, $K(\psi)$, are required. The aim of this investigation was to measure the parameters for input to the model, to assess the variability of these parameters and to determine whether or not parameters should be used for each domain independently. A Guelph permeameter was used to obtain K_s and α , which describes the slope of the logarithmic unsaturated hydraulic conductivity function ($K(\psi) = K_s \exp(\alpha\psi)$). Measurements at four depths were carried out. Domains were chosen according to land use — arable land (including temporary grass), permanent grassland and woodland. Both K_s and α were found to be log-normally distributed. The K_s values were highest in the grassland and lowest in arable land. The difference between means was significant for K_s but not for α in the three domains. High values of K_s were attributed to preferential flow along slate faces and root-induced channels in the grassland and woodland, respectively, as indicated by dye tests. Finally, it was concluded that, as K_s and $K(\psi)$ differ significantly between domains, different hydrological behaviour would be expected. It is therefore recommended that different parameter sets be used for modelling the individual domains.

Introduction

Most hydrological, soil water balance and chemical transport models require saturated and unsaturated soil hydraulic conductivities to solve the

* Corresponding author.

transport equations under saturated and unsaturated conditions. Preferential flow paths and spatial variability of hydraulic conductivity greatly influence the performance of these models in predicting transport of water and chemicals.

The objectives of this research were as follows: to derive appropriate data for hydraulic conductivity to use as input to the European Hydrological System (Système Hydrologique Européen — SHE) (Abbott et al., 1986a,b); to investigate the variability of saturated and unsaturated hydraulic conductivity between the three land use domains — arable land, permanent grassland and woodland — found in a small catchment; to decide whether or not independent parameter sets should be used for the three domains; to assess the importance of preferential flow in the soils of the catchment.

The Guelph permeameter

Methods for obtaining soil hydraulic properties have received considerable attention over the past three decades. Several field and laboratory methods and predictive models have been developed. Ragab and Cooper (1990) classified these methods according to their application conditions, accuracy, effective costs, advantages and limitations and concluded that, under field conditions and in the absence of a shallow water table, the Guelph permeameter was the most useful. It causes minimal disturbance, is easily employed, portable and durable, allows rapid field calculation of saturated hydraulic conductivity, is accurate and allows several replicate measurements to be made in a single day. The method yields a combination of vertical and horizontal saturated conductivity representing the actual situation in the field, it is suitable for various soil types and, in addition to the measured hydraulic conductivity, the method provides an estimate of the unsaturated hydraulic function. The method or its principles have been used by Talsma and Hallam (1980), Stephens et al. (1987), Bell and Schofield (1990), Heinen and Raats (1990), Jarbo and Fritton (1990), Logsdon et al. (1990).

The Guelph permeameter method, as described by Reynolds et al. (1983, 1985), Reynolds and Elrick (1985, 1987, 1990) and Elrick et al. (1990), involves augering a vertical borehole and determining the steady discharge of water when the level in the well is held constant. Reynolds et al. (1985) proposed sequential measurements of steady-state flux, Q , out of an auger hole at two constant depths of ponding in the hole, H , and the use of Richards' equation to calculate simultaneously the field saturated hydraulic conductivity, K_{fs} , from the two measurements. Reynolds and Elrick (1985) and Reynolds et al. (1985) presented an analytical solution for the case of steady

recharge from a cylindrical, uncased well into uniform, unsaturated soil as follows:

$$2\pi H^2 K_{fs} + C\pi a^2 K_{fs} + 2\pi H\phi_m = CQ \quad (1)$$

where a is the well radius, ϕ_m is the matric flux potential and C is a dimensionless proportionality parameter, primarily dependent on the H/a ratio. The first term on the left of Eq. (1) represents the ‘hydraulic push’ of water into the soil as a result of the hydrostatic pressure of the water in the well, the second term represents the ‘gravitational pull’ of water out through the bottom of the well, and the third term represents the ‘matric pull’ of water out of the well as a result of capillary forces in the soil. The first two terms of Eq. (1) may be thought of as the ‘field saturated’ component of flow out of the well and the third as the unsaturated flow component. Equation (1) may be solved for K_{fs} and ϕ_m using either a simultaneous equations or a least-squares approach, dependent on the number of pairs of values of H and Q which have been measured. Under field conditions, a completely saturated condition is rarely reached, owing to entrapped air in the micropores. A maximum attainable value of saturated hydraulic conductivity, K_{fs} , is therefore obtained.

The matric flux potential, ϕ_m , was defined by Gardner (1958) as

$$\phi_m = \int_{\psi_i}^0 K(\psi) d\psi, \quad \psi_i \leq \psi < 0 \quad (2)$$

where ψ is the soil (pore) water pressure head, ψ_i is the initial pressure head in the soil surrounding the well (ψ_i is assumed constant and $-\infty \leq \psi_i \leq 0$), and $K(\psi)$ is the hydraulic conductivity — pressure head relationship for infiltration. ϕ_m decreases from a soil-dependent maximum value to zero as ψ_i increases from $-\infty$ (dry soil) to zero (field-saturated soil).

A popular empirical representation of the $K(\psi)$ relationship was given by Gardner (1958):

$$K(\psi) = K_{fs} \exp(\alpha\psi), \quad 0 < \alpha < \infty, \quad \psi \leq 0 \quad (3)$$

where α is the slope of $\ln K$ vs. ψ . An exponential relationship of the form given in Eq. (3) has been found to give a good approximation to the actual curve measured for a wide range of soils over a significant range of water status (Philip, 1968). Substituting Eq. (3) into Eq. (2) and integrating allows capillarity (the ratio of ϕ_m to K_{fs}) to be expressed in terms of α and ψ_i :

$$\frac{\phi_m}{K_{fs}} = \frac{[1 - \exp(\alpha\psi_i)]}{\alpha} \quad (4)$$

For most soils at field capacity and drier, $\exp(\alpha\psi_i)$ is small relative to unity, and consequently Eq. (4) can be approximated by

$$\alpha^{-1} = \frac{\phi_m}{K_{fs}} \quad (5)$$

As a result, α^{-1} often closely approximates the capillarity of an unsaturated soil. Philip (1985) referred to α^{-1} as 'sorptive length'. It is a measure of a soil's ability to absorb water. The larger this value, the greater the sorptive ability or 'capillarity'. It increases from sand (coarse texture) to clay (fine texture). However, soil structure in the form of macropores can counteract this trend (Reynolds and Elrick, 1987). α^{-1} is also characterised by its hysteretic behaviour and so depends on whether the soil is draining or wetting. The latter condition is relevant to the Guelph permeameter method.

Equations (1)–(4) reveal that the balance between the field saturated and capillary components of flow out of the well depends on H , a , α and ψ_i . The relative importance of the field-saturated component of flow increases when any of well radius (a), depth of ponding in the well (H), coarseness of soil texture (decreasing α^{-1}) or initial soil wetness (θ_i and ψ_i) increases. Conversely, the capillary component increases with decreasing H and a , with finer soil texture (increasing α^{-1}) and with decreasing θ_i and ψ_i .

Sites

Slapton Wood catchment was chosen for this study. A large number of hydrological and nitrate balance studies have been conducted in the catchment over a period of 20 years (e.g. Troake and Walling, 1973; Trudgill, 1983; Burt et al., 1983; Coles, 1985). The catchment is located 10 km south-west of Dartmouth, Devon, UK. Its total area is 0.94 km² and it is underlain by slates and shales of Lower Devonian age. Figure 1 gives a map of the catchment showing land use in 1990.

The catchment has four major land use domains, which in summer 1990, when the experiments reported here were conducted, comprised the following percentages of the catchment area: temporary grassland, 39%; arable land, 20%; permanent grassland, 24%; woodland, 14%; the rest is used for roads, buildings, etc. The arable and temporary grass areas change from one to the other according to crop rotations from year to year. Accordingly, these two land uses were considered to be one domain. The domains identified are therefore arable land (including temporary grass), permanent grassland and woodland.

Topographically, the catchment is dominated by gently sloping ground

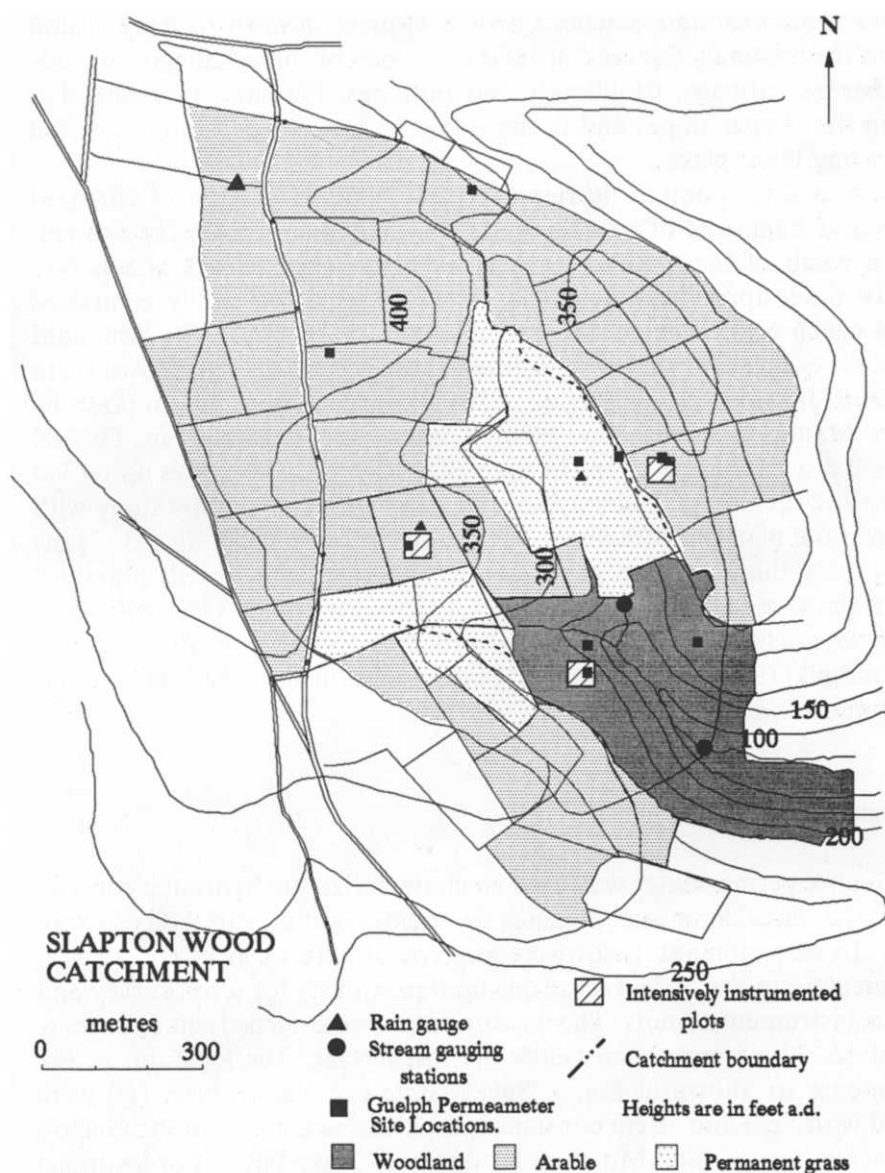


Fig. 1. Map of Slapton Wood catchment, showing the location of the Guelph permeameter measurements.

(about 5°) above the 90 m contour, which affords excellent mixed farming, with below this much steeper valley sides, with slopes up to 25°. Vegetation in the woodland consists mainly of sweet chestnut (*Castanea sativa*), but also oak (*Quercus hybrid petraearobur*) and beech (*Fagus sylvatica*). The ground flora consists of bluebell (*Endymion scilla non-scripta*), wood anemone

(*Anemone nemorosa*) and butcher's broom (*Ruscus aculeatus*). Agricultural crops, on the relatively flat land around the flanks of the catchment, include wheat, barley, cabbage, cauliflower and potatoes. Permanent grassland is found on the steeper slopes and in the valley bottom, where light sheep and cattle grazing takes place.

The soil is acid, poor in nutrients, with a high proportion of silt-sized particles and fragments of weathered slates. It has a strong red–brown colour, as a result of the presence of oxidized iron. The bedrock is a Lower Devonian slate, approximately 390 million years old, originally composed of muds which have been compressed and metamorphosed. The dominant minerals are quartz and kaolinite. Soil type ranges from an acid brown earth of the Denbigh series (Findlay et al., 1984) towards a more brown podzolic soil of the Manod series (Bradley, 1976). A true podzol is not present. The soil profile is 0.6–1.0 m deep, except in the valley bottom, where it is up to 3 m deep. The soil texture is classified as a silty clay loam. It is slightly stony with subangular and platy slate fragments. The sand (2 mm–63 μ m), silt (63–2 μ m) and clay (less than 2 μ m) percentages according to the British Standard classification were 30%, 45% and 25% for the arable land, 33%, 43%, and 24% for the grassland, and 33%, 41% and 26% for the woodland, respectively. Trudgill (1983) and Coles (1985) have given further details of the soils of the Slapton Wood catchment.

Experimental methods

The Guelph permeameter was used to characterize the hydraulic conductivity of the three land use domains including spatial variability within domains. In each domain, holes were augered at various locations, chosen at random close to the soil pit locations used previously for a soil survey and within the instrumented plots. The measurements were carried out at average depths of 15, 30, 45 and 60 cm below the soil surface. The locations of the measurements are shown in Fig. 1. Steady-state percolation rates (Q) were measured with 5 cm and 10 cm constant head levels, according to the Guelph Permeameter manual (Soil Moisture Equipment Corp., 1987). For locations with extensive heterogeneity, smaller head (H) levels were used, as recommended by Reynolds and Elrick (1986). Calculations were made following the instructions outlined by Reynolds and Elrick (1986) and the instrument manual (Soil Moisture Equipment Corp., 1987).

Flow pathways were investigated by a staining test. Methylene Blue solution was introduced using the Guelph permeameter in a number of augered

holes in the three land use domains. At the start, the permeameter was filled completely with the dye solution. The whole volume of the solution was allowed to percolate through the auger hole, followed by excavation of a soil pit, the face of which was cut back progressively towards the auger hole.

Results and discussion

The values of the field saturated hydraulic conductivity and the matric flux potential were obtained by simultaneous solution of Eq. (1) for two constant heads, H , and the steady-state rates, Q , obtained from the field measurements. The logarithmic slope of the unsaturated hydraulic conductivity–soil matric suction relationship, α , was obtained according to Eq. (5). Figures 2, 3 and 4 show the saturated hydraulic conductivity, matric flux potential and α value for the arable land, grassland and woodland at various depths. Table 1 shows the number of observations, arithmetic and geometric mean, variance, standard deviation, standard error and the coefficient of variation of K_{fs} , ϕ_m and α for each domain at various soil depths.

The mean saturated hydraulic conductivity at each depth was found to be in the range of 25–31 cm day⁻¹, 79–252 cm day⁻¹ and 69–123 cm day⁻¹ for arable land, grassland and woodland, respectively. The coefficient of variation (CV) ranged from 37 to 60%, 50 to 75% and 59 to 131%, respectively, for these domains. The arable land (including temporary grass) had the lowest and the woodland had the highest coefficient of variation. Lee et al. (1985) reported Guelph permeameter measurements for various soil types. They reported K_{fs} values of 330 cm day⁻¹, 11.40 cm day⁻¹, 11.23 cm day⁻¹ and 0.73 cm day⁻¹ for loamy sand, very fine sandy loam, silt loam and clay, respectively. The CV values ranged between 75 and 627%. Reynolds and Elrick (1985) obtained a mean K_{fs} value of 28.94 cm day⁻¹ for loamy soil under corn. The CV ranged between 40 and 120%, which is similar to the values presented here. Wilson et al. (1989) used the same method on two forested catchments dominated by weathered clay subsoils. The average K_{fs} values were 1831.68 and 478.66 cm day⁻¹, with CV ranges between 268 and 338%. The CVs found in this study, therefore, are towards the lower end of those found by other workers.

The matric flux potential values were in the range from 62–280 cm² day⁻¹, 83–1699 cm² day⁻¹ and 416–511 cm² day⁻¹ for the arable land, grassland and woodland, respectively. The grassland had the highest values at 45 and 60 cm depths and the lowest values at 15 and 30 cm depths, whereas woodland showed higher values compared with the arable land at all depths. The CV ranged from 93 to 141%, 49 to 89% and 77 to 174% for arable land, grassland

SATURATED HYDRAULIC CONDUCTIVITY OF THE SLAPTON WOOD CATCHMENT

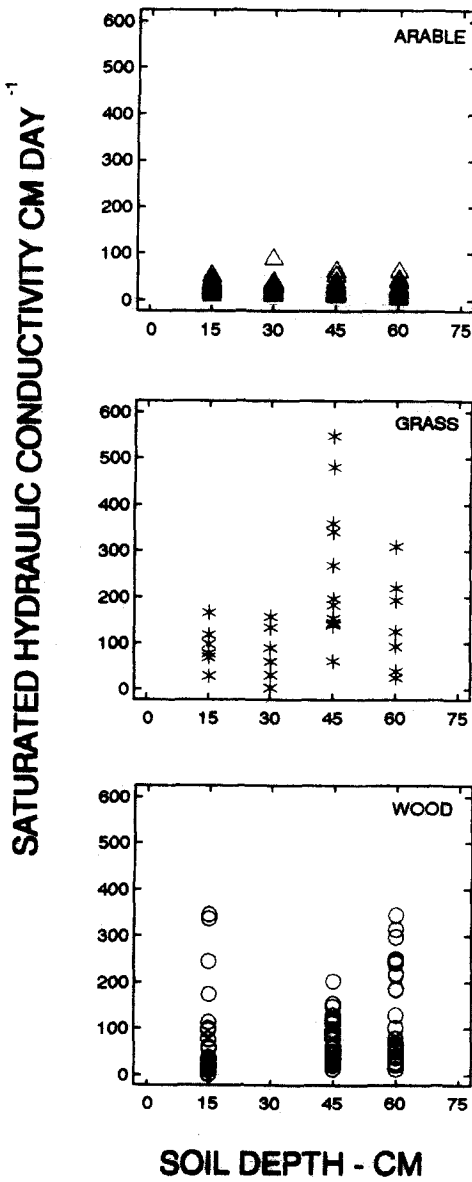


Fig. 2. Saturated hydraulic conductivity of the three land use domains of the Slapton Wood catchment.

MATRIC FLUX POTENTIAL OF THE SLAPTON CATCHMENT

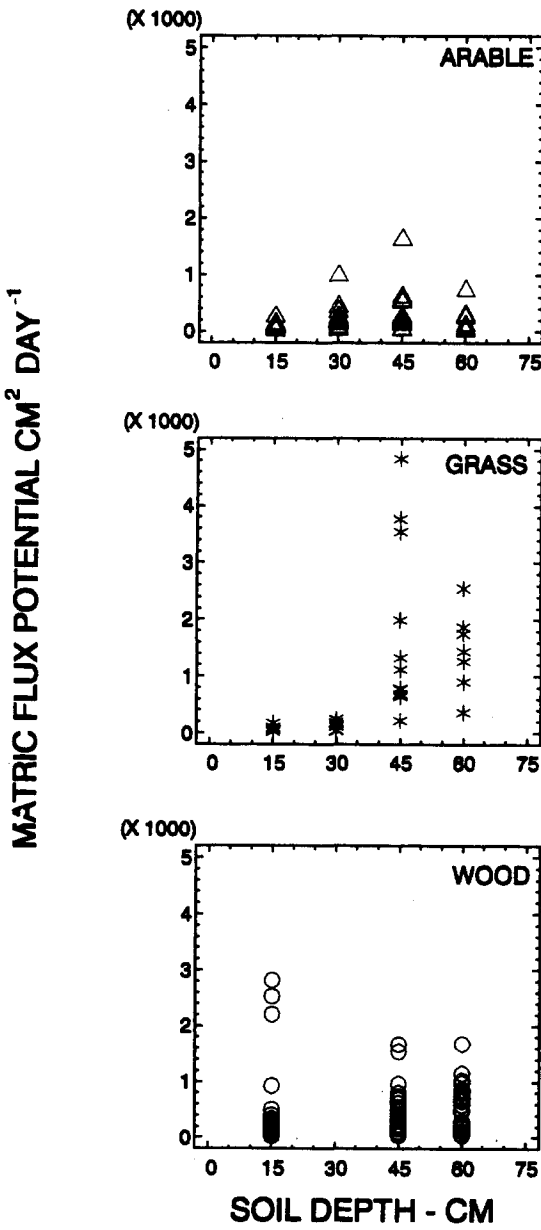


Fig. 3. Matric flux potential of the three land use domains of the Slapton Wood catchment.

SLOPE OF THE UNSATURATED HYDRAULIC CONDUCTIVITY FUNCTION OF THE SLAPTON CATCHMENT

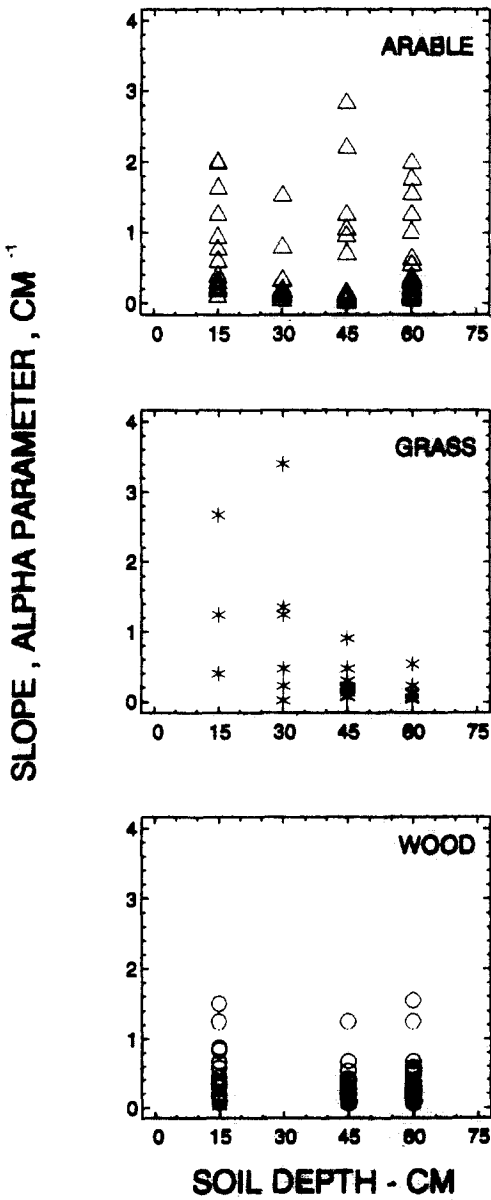


Fig. 4. Slope of the unsaturated hydraulic conductivity function of the three land use domains of the Slapton Wood catchment.

and woodland, respectively. The results show that the matric flux potential has, in general, a tendency to increase with depth, except for the 60 cm depth of arable land.

The α values were in the range $0.28\text{--}0.89\text{ cm}^{-1}$, $0.15\text{--}1.34\text{ cm}^{-1}$ and $0.36\text{--}0.37\text{ cm}^{-1}$ for the arable land, grassland and woodland, respectively, and the coefficients of variation range from 70 to 147%, 55 to 120% and 94 to 108%, respectively. Clothier and Scotter (1982) obtained $\alpha = 0.3052\text{ cm}^{-1}$ for fine sand. Philip (1985) reported values of 0.05 cm^{-1} and 0.002 cm^{-1} for coarse- and fine-grained soils, respectively, which had been air-dried and sieved. For sand, loam and clay, Reynolds and Elrick (1985, 1987) found values of 0.25 cm^{-1} , 0.0336 cm^{-1} and 0.0010 cm^{-1} , respectively, and indicated that for sand α may vary between 0.10 and 0.50 cm^{-1} . Scotter et al. (1982) found the following ranges: $0.50\text{--}0.92\text{ cm}^{-1}$ for sandy loam, $0.04\text{--}0.13\text{ cm}^{-1}$ for fine sandy loam and about 0.02 cm^{-1} for silt loam to clay loam. For a wide range of soil types, from sand to clay, α ranges from 0.35 to 0.06 cm^{-1} , according to Talsma (1987). White and Sully (1987) found α to range from 0.71 cm^{-1} for gravelly loam fine sand to 0.05 cm^{-1} for clay. They stated that, for soils with high saturated conductivity with a high percentage of sand or gravel, typical field values cover a range from 0.10 to 1.00 cm^{-1} . Most of the α values reported here are within this range.

The frequency distributions of K_{ks} , ϕ_m and α for all the results obtained were fitted to both a normal and a log-normal distribution. The results indicate that the K_{fs} , ϕ_m and α values are all much closer to being log-normally distributed than normally distributed, as shown in Fig. 5. This trend has been reported by many workers, e.g. Talsma and Hallam (1980), Reynolds and Elrick (1985), Lee et al. (1985), Wilson et al. (1989), Espeby (1990), and Peck et al. (1990). Consequently, the geometric mean was considered to be the best estimate for the mean, rather than the arithmetic mean (Campbell, 1985).

The geometric mean of K_{fs} , ϕ_m and α for the three land use domains at various depths are shown in Fig. 6. The figure shows that the grassland has the highest mean values of saturated hydraulic conductivity and the arable land has the lowest. Moreover, the highest mean value was found at 45 cm depth in the grassland. The figure shows also that the matric flux potential is greater at 45 and 60 cm than at 15 and 30 cm depths in both grassland and woodland, though not in arable land. The largest values were found in the grass at 45 and 60 cm depths. By contrast, alpha values are larger at the 15 and 30 cm depth and tend to decrease with depth. The grassland had the highest value at the 15 and 30 cm depths. The geometric mean of α for the grassland and arable land had a general tendency to decrease with depth, whereas for woodland it was relatively constant.

Statistical analysis was carried out to test the significance of the differences

Table 1

Summary statistics of saturated hydraulic conductivity, matric flux potential and slope of the unsaturated hydraulic conductivity function at four depths (in cm) for the three land use domains of Slapton Wood catchment

Arable														
15 cm			30 cm			45 cm			60 cm					
K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α
No. of observations	19	19	19	15	15	15	17	17	17	22	22	22	22	22
Average	31.28	61.79	0.89	30.56	227.84	0.28	27.15	279.89	0.57	24.59	116.54	0.61	116.54	0.61
Geometric mean	28.83	45.15	0.64	27.05	158.38	0.17	23.68	132.89	0.18	20.75	62.92	0.33	62.92	0.33
Variance	132.37	3317.59	0.38	331.07	54 339.80	0.15	215.60	155 464.00	0.70	192.51	26 436.40	0.38	26 436.40	0.38
SD	11.51	57.60	0.62	18.20	233.11	0.39	14.68	394.29	0.84	13.88	162.59	0.62	162.59	0.62
SE	2.64	13.21	0.14	4.70	60.19	0.10	3.56	95.63	0.21	2.96	34.67	0.13	34.67	0.13
CV (%)	36.80	93.22	69.66	59.56	102.31	139.29	54.07	140.87	147.37	56.45	139.51	101.64	139.51	101.64

Grass														
15 cm			30 cm			45 cm			60 cm					
K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α
No. of observations	6	6	6	6	6	6	12	12	12	7	7	7	7	7
Average	93.64	83.38	1.34	79.29	131.20	1.11	252.01	1698.60	0.25	143.63	1448.94	0.15	1448.94	0.15
Geometric mean	82.57	70.76	1.17	46.42	106.84	0.44	212.91	1184.28	0.18	106.61	1257.89	0.09	1257.89	0.09
Variance	2166.65	2621.53	0.54	3560.32	6347.89	1.54	22 750.20	2 290 000.00	0.06	10 516.30	498 666.00	0.03	498 666.00	0.03
SD	46.55	51.20	0.73	59.67	79.67	1.24	150.83	1513.69	0.23	102.55	706.16	0.18	706.16	0.18
SE	19.00	20.90	0.30	24.36	32.53	0.51	43.54	436.97	0.07	38.76	266.90	0.07	266.90	0.07
CV (%)	49.71	61.41	54.48	75.26	60.72	111.71	59.85	89.11	92.00	71.40	48.74	120.00	48.74	120.00

Wood											
15 cm				45 cm				60 cm			
	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m
No. of observations	31	31	31	30	30	30	30	30	30	30	30
Average	69.04	415.94	0.37	77.90	442.79	0.36	123.15	511.36	0.36	0.36	0.36
Geometric mean	34.62	1172.88	0.21	65.16	304.61	0.21	88.80	362.33	0.25	0.25	0.25
Variance	8171.45	527 380.00	0.15	2087.16	156 194.00	0.15	9758.02	156 159.00	0.11	0.11	0.11
SD	90.40	726.21	0.39	45.69	395.21	0.39	98.78	395.17	0.34	0.34	0.34
SE	16.24	130.43	0.007	8.34	72.16	0.07	18.04	72.15	0.06	0.06	0.06
CV (%)	130.94	174.60	105.41	58.65	89.25	108.33	80.21	77.28	94.44	94.44	94.44

K_{fs} — Saturated hydraulic conductivity (cm day^{-1}); ϕ_m — matric flux potential ($\text{cm}^2 \text{day}^{-1}$); α — slope of the unsaturated hydraulic conductivity function, $K(\psi)$ (cm^{-1}).

FREQUENCY DISTRIBUTION : GUELPH PERMEAMETER METHOD.

RELATIVE FREQUENCY

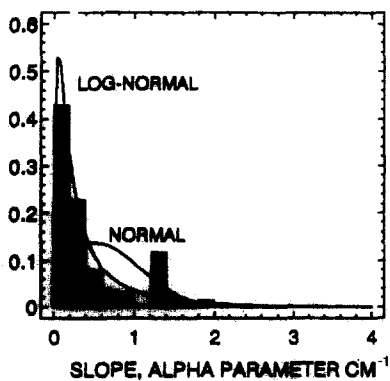
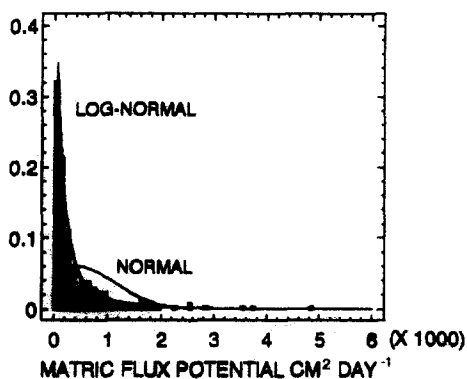
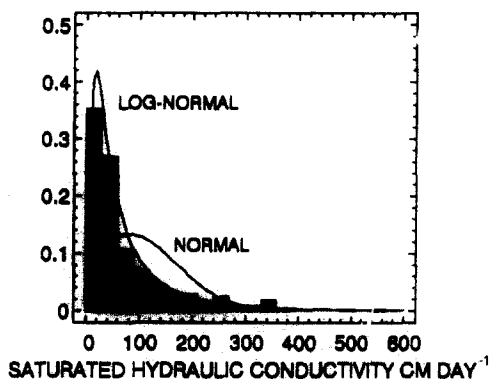


Fig. 5. Frequency distribution of the saturated hydraulic conductivity, matric flux potential and slope of the unsaturated hydraulic conductivity function of the Slapton Wood catchment. The solid lines show best-fit normal and log-normal distributions for these data.

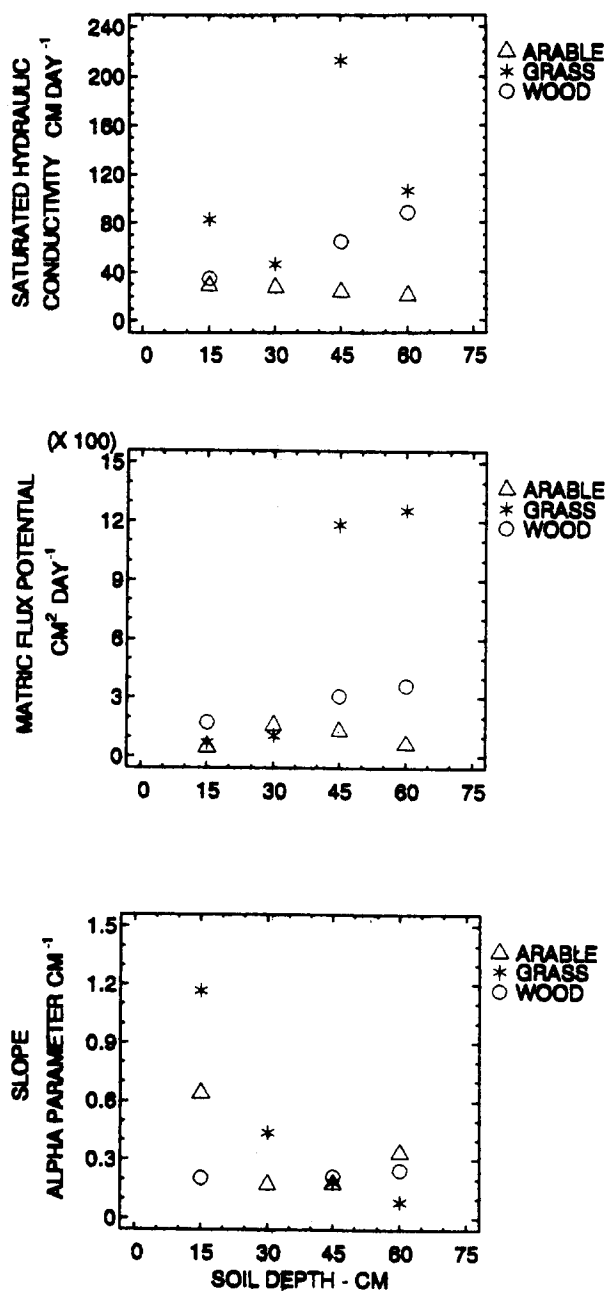


Fig. 6. Geometric mean of saturated hydraulic conductivity (top), matric flux potential (middle) and slope of the unsaturated hydraulic conductivity function (bottom) of the three land use domains of the Slapton Wood catchment.

between means of K_{fs} , ϕ_m and α of the three domains at a 95% significance level (Table 2). The analysis indicated that the K_{fs} of the arable land differs significantly from K_{fs} of the grassland and woodland at three depths. The ϕ_m of arable land differs significantly at all depths from the ϕ_m of the woodland and at two depths from the ϕ_m of the grassland. By contrast, the α parameter did not differ significantly between the arable land and either the grassland or woodland except at one depth in each, suggesting that differences in ϕ_m are due primarily to differences in K_{fs} , rather than in α .

When comparing the grassland and woodland, it was found that K_{fs} differs significantly at the 45 cm depth only, although no comparison could be made for the 30 cm depth. ϕ_m differs significantly at all depths, whereas the α parameter is significantly different at the 15 and 60 cm depth. Variations of K_{fs} and α with depth within each domain were significant in grassland and woodland but not in arable land. On a catchment scale (all domains combined), K_{fs} variation with depth was significant, but variation of α was not significant. When the three domains were compared, using all the depths of measurement (Table 2), there were significant differences of K_{fs} and ϕ_m between the three domains, but not of α .

The summary statistics for the three domains, regardless of the depth of measurement, are given in Table 3. This table shows that the grassland had the highest values of both K_{fs} and ϕ_m , the arable land had the lowest values of both K_{fs} and ϕ_m and the woodland had intermediate values. The geometric means of the α values are only slightly different between domains and within one standard error of one another. Moreover, there was no correlation between K_{fs} and α in the three domains.

Table 2

Significance test between means of saturated hydraulic conductivity, matric flux potential and slope of the unsaturated hydraulic conductivity function of the three land use domains of Slapton Wood catchment

Depth (cm)	Arable vs. grass			Arable vs. wood			Grass vs. wood		
	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α
15	*	NS	NS	NS	*	*	NS	*	*
30	NS	NS	NS						
45	*	*	NS	*	*	NS	*	*	NS
60	*	*	*	*	*	NS	NS	*	*
0–60	*	*	NS	*	*	NS	*	*	NS

*Significant at 95% confidence level.

NS — Not significant at 95% confidence level.

Table 3

Summary statistics of saturated hydraulic conductivity, matric flux potential and the slope of the unsaturated hydraulic conductivity function of the three land use domains of Slapton Wood catchment

	Arable			Grass			Wood		
	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α	K_{fs}	ϕ_m	α
No. of observations	73	73	73	31	31	31	91	91	91
Average	28.16	163.20	0.61	163.46	1026.23	0.61	89.80	456.25	0.36
Geometric mean	24.61	83.03	0.30	112.90	436.84	0.26	58.17	267.37	0.22
Variance	209.42	61118.40	0.43	17019.50	1500000.00	0.63	7106.98	278073.00	0.14
SD	14.47	247.22	0.66	130.46	1225.52	0.80	84.30	527.33	0.37
SE	1.69	28.94	0.08	23.43	220.11	0.14	8.84	55.28	0.04
CV (%)	51.39	151.48	108.20	79.81	119.42	131.48	93.88	115.58	102.78

The unsaturated hydraulic conductivity function was calculated for each domain from K_{fs} and α . This leads to the following equations:

for arable land:

$$K(\psi) = 24.61 \exp(0.30\psi) \quad (6a)$$

for grassland:

$$K(\psi) = 112.90 \exp(0.26\psi) \quad (6b)$$

for woodland:

$$K(\psi) = 58.17 \exp(0.22\psi) \quad (6c)$$

where $K(\psi)$ is expressed in centimetres per day and ψ in centimetres (negative values). The three functions are shown in Fig. 7. The function gives a straight line when $\log K(\psi)$ is plotted against ψ , with slope α and intercept K_{fs} . As the differences between means of K_{fs} were significant, the differences between means of $K(\psi)$ will also be significant between domains.

The association of soil hydraulic parameters with land use was made because this was considered to be the most suitable division of the catchment hydrologically. The hydrological behaviour of soils in the catchment is

UNSATURATED HYDRAULIC CONDUCTIVITY OF SLAPTON CATCHMENT OBTAINED FROM THE GUELPH PERMEAMETER METHOD.

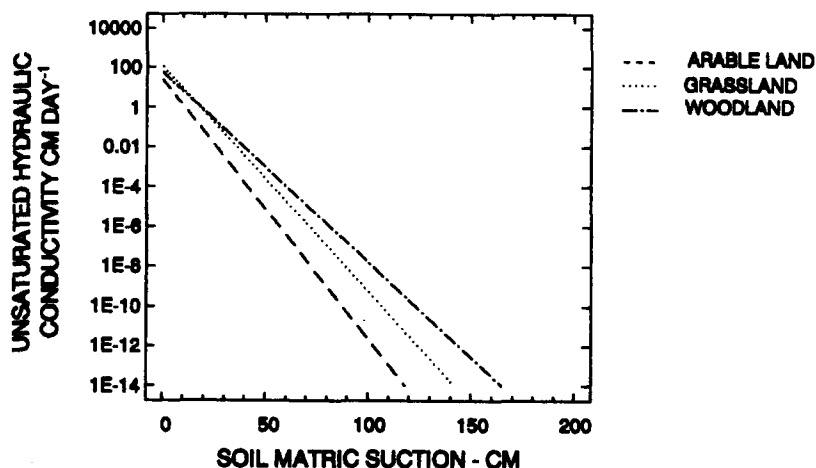


Fig. 7. Unsaturated hydraulic conductivity as a function of matric flux potential for the three land use domains of the Slapton Wood catchment.

expected to be similar, as there are no clear differences in particle size or development of, for example, impeding horizons. However, there is a strong association of land use with soil type, with Manod series soils (Bradley, 1976) prevailing in the woodland, Denbigh soils (Findlay et al., 1984) in the temporary grassland and arable land, and Manod, Denbigh and Powys (a shallow phase of Denbigh) soil on the steep slopes of permanent grassland, with some Ravenstock soil in the valley bottoms. With the information available, it is not, therefore, possible to identify the differences found as being caused by land use. They may equally well be a reflection of soil type.

The importance of preferential pathways

The staining test, using Methylene Blue, allows direct visual observation and assessment of preferential pathways. Methylene Blue has a large molecular structure with a low diffusion coefficient and is strongly adsorbed by most soil materials; thus, only the pathways conducting rapidly flowing water absorb the dye. Figure 8 shows illustrative results of the test on the arable land (at present under temporary grass), permanent grassland and woodland. For the arable land (under temporary grass) the wet bulb approximates the classical spheroid shape. The grass roots appeared stained, suggesting that root-induced macropores conducted the dye. The photograph for permanent grassland shows that the soil profile appears to divide into two distinct layers; the top layer is relatively compact whereas the lower layer contains large slates with clear large cleavages. The bulb shape observed in the arable land is absent from this profile. Instead, the dye appeared at the slate faces. The soil profile in the woodland is relatively homogeneous and contains roots of a larger size. The photograph shows that the stained parts are actually root-induced channels.

Accordingly, one may attribute the relatively high values of K_{fs} of both grassland and woodland to the presence of preferential flow along slate faces and root-induced channels, respectively. Similar results were obtained by Wilson et al. (1989). In their study on two catchments, they found that the greater K_{fs} could be attributed to the fractured nature of the saprolite, which was apparently more hydrologically continuous in one catchment than the macropores developed predominantly by deep rooting at the other.

Conclusion

Physically based hydrological models (of which SHE is an example) require information on saturated and unsaturated hydraulic conductivity of soils. These two characteristics are known to be highly variable. Land use is an

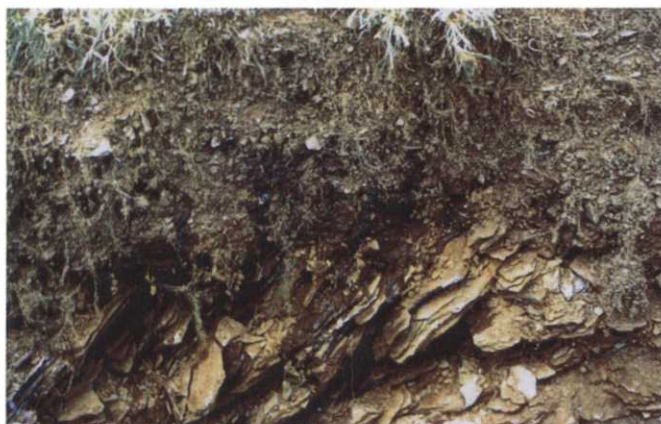
**ARABLE****PERMANENT
GRASS****WOODLAND**

Fig. 8. Tracing flow paths by using Methylene Blue dye in arable land (temporary grass), permanent grassland and woodland.

important factor which affects their value. Model performance is sensitive to these characteristics and hence realistic simulation can be expected only if realistic values are used. This study revealed that the three land use domains of Slapton Wood catchment have significantly different conductivity characteristics. Therefore different hydrological behaviour would be expected in each domain and independent parameter sets should be used when applying a physically based model such as SHE to the catchment.

Acknowledgments

This study was funded by UK Nirex Ltd. as part of its programme of research into the safety of the deep underground disposal of low- and intermediate-level radioactive waste. The Slapton Wood area is not being considered for such disposal. Thanks are due to J. and P. Cocks, and A.R. and P. Aubrey, and to the Slapton Ley Field Centre for permission to carry out measurements on their land and for facilitating the project.

References

- Abbott, M.B., Bathurst, J.C., Cunge, J.A., O'Connell, P.E. and Rasmussen, J., 1986a. An introduction to the European Hydrological System — *Système Hydrologique Européen* SHE, 1: History and philosophy of a physically-based, distributed modelling system. *J. Hydrol.*, 87: 45–59.
- Abbott, M.B., Bathurst, J.C., Cunge, J.A., O'Connell, P.E. and Rasmussen, J., 1986b. An introduction to the European Hydrological System — *Système Hydrologique Européen* SHE, 2: Structure of a physically-based, distributed modelling system. *J. Hydrol.*, 87: 61–77.
- Bell, R.W. and Schofield, N.J., 1990. Design and application of a constant head well permeameter for shallow high saturated hydraulic conductivity soils. *Hydrol. Processes*, 4: 327–342.
- Bradley, R.I., 1976. Soils in Dyfed III: Sheet SN 13 (Eglwysrw). *Soil Survey Rec.* 38. Soil Survey of England and Wales, Harpenden.
- Burt, T.P., Butcher, D.P., Coles, N. and Thomas, A.D., 1983. The natural history of Slapton Ley nature reserve XV: Hydrological processes in the Slapton Wood catchment. *Field Stud.*, 5: 731–752.
- Campbell, G.S., 1985. *Soil Physics with BASIC. Transport Models for Soil-Plant Systems.* Developments in Soil Science, 14. Elsevier, Amsterdam, pp. 60–72.
- Clothier, B.E. and Scotter, D.R., 1982. Constant-flux infiltration from a hemispherical cavity. *Soil Sci. Soc. Am. J.*, 46: 696–700.
- Coles, N., 1985. Nitrate movement by preferential flow in a weakly structured soil. Ph.D. Thesis, University of Sheffield.
- Elrick, D.E., Reynolds, W.D., Geering, H.R. and Tan, K.A., 1990. Estimating steady infiltration rate times for infiltrometers and permeameters. *Water Resour. Res.*, 26: 759–769.
- Espeby, B., 1990. An analysis of saturated hydraulic conductivity in a forested glacial till slope. *Soil Sci.*, 150: 485–494.

- Findlay, D.C., Colborne, G.J.N., Cope, D.W., Harrod, T.R., Hogan, D.V. and Staines, S.J., 1984. Soils and their use in South West England. *Soil Surv. England and Wales*, Harpenden, Bull. 14, pp. 145–151.
- Gardner, W.R., 1958. Some steady-state solutions of the unsaturated moisture flow equation with application to evaporation from a water table. *Soil Sci.*, 85: 228–232.
- Heinen, M. and Raats, P.A.C., 1990. Evaluation of two models describing the steady discharge from a constant head well permeameter into unsaturated soil. *Soil Sci.*, 150: 401–412.
- Jarbo, J.D. and Fritton, D.D., 1990. Simulation of water flow from a percolation test hole in layered soil. *Soil Sci. Soc. Am. J.*, 54: 1214–1218.
- Lee, D.M., Reynolds, W.D., Elrick, D.E. and Clothier, B.E., 1985. A comparison of three field methods for measuring hydraulic conductivity. *Can. J. Soil Sci.*, 65: 563–573.
- Logsdon, S.D., Allmaras, R.R., Wu, L., Swan, J.B. and Randall, G.W., 1990. Macroporosity and its relation to saturated hydraulic conductivity under different tillage practices. *Soil Sci. Soc. Am. J.*, 54: 1096–1101.
- Peck, A.J., Yendle, P.A. and Batini, F.E., 1990. Hydraulic conductivity of deeply weathered materials in the Darling Range, Western Australia. *Aust. J. Soil Res.*, 18: 129–138.
- Philip, J.R., 1968. Steady infiltration from buried point sources and spherical cavities. *Water Resour. Res.*, 4: 1039–1047.
- Philip, J.R., 1985. Approximate analysis of the borehole permeameter in unsaturated soil. *Water Resour. Res.*, 21: 1025–1033.
- Ragab, R. and Cooper, J.D., 1990. Obtaining soil hydraulic properties from field, laboratory and predictive methods. Nirex Safety Studies. NSS/R226. UK Nirex. Didcot. 83 pp.
- Reynolds, W.D. and Elrick, D.E., 1985. In situ measurements of field-saturated hydraulic conductivity, sorptivity, and the α -parameter using the Guelph Permeameter. *Soil Sci.*, 140: 292–302.
- Reynolds, W.D. and Elrick, D.E., 1986. A method for simultaneous in situ measurement in the vadose zone of field-saturated hydraulic conductivity, sorptivity, and the conductivity–pressure head relationship. *Ground Water Monit. Rev.*, 6: 84–95.
- Reynolds, W.D. and Elrick, D.E., 1987. A laboratory and numerical assessment of the Guelph Permeameter method. *Soil Sci.*, 144: 282–299.
- Reynolds, W.D. and Elrick, D.E., 1990. Pondered infiltration from a single ring: I. Analysis of steady flow. *Soil Sci. Soc. Am. J.*, 54: 1233–1241.
- Reynolds, W.D., Elrick, D.E. and Topp, G.C., 1983. A reexamination of the constant head well permeameter method for measuring saturated hydraulic conductivity above the water table. *Soil Sci.*, 136: 250–268.
- Reynolds, W.D., Elrick, D.E. and Clothier, B.E., 1985. The constant head well permeameter: effect of unsaturated flow. *Soil Sci.*, 139: 172–180.
- Scotter, D.R., Clothier, B.E. and Harper, E.R., 1982. Measuring saturated hydraulic conductivity and sorptivity using twin rings. *Aust. J. Soil Res.*, 20: 295–304.
- Soil Moisture Equipment Corp., 1987. Guelph Permeameter. Model 2800KI. Operating and Instruction Manual. Soil Moisture Equipment Corp., Santa Barbara, CA. 28 pp.
- Stephens, D.B., Lambert, D. and Watson, D., 1987. Regression models for hydraulic conductivity and field test of the borehole permeameter. *Water Resour. Res.*, 23: 2207–2214.
- Talsma, T., 1987. Re-evaluation of the well permeameter as a field method for measuring hydraulic conductivity. *Aust. J. Soil Res.*, 25: 361–368.
- Talsma, T. and Hallam, P.M., 1980. Hydraulic conductivity measurement of forest catchments. *Aust. J. Soil Res.*, 30: 139–148.

- Troake, R.P. and Walling, D.E., 1973. The natural history of Slapton Ley nature reserve VII: The hydrology of the Slapton Wood stream. *Field Stud.*, 3: 719–740.
- Trudgill, S.T., 1983. The natural history of Slapton Ley nature reserve, XVI: The soils of Slapton Wood. *Field Stud.*, 5: 833–840.
- White, I. and Sully, M.J., 1987. Macroscopic and microscopic capillary length and time scales from field infiltration. *Water Resour. Res.*, 23: 1514–1522.
- Wilson, G.V., Alfonsi, J.M. and Jardine, P.M., 1989. Spatial variability of saturated hydraulic conductivity of the subsoil of two forested watersheds. *Soil Sci. Soc. Am. J.*, 53: 679–685.